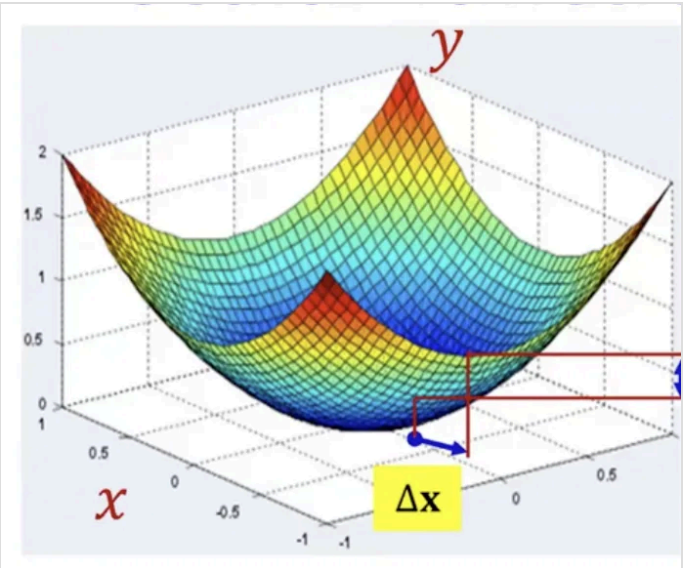
Lec4. Training Part II

function optimization



$$\Delta y = \nabla_{\mathbf{x}} y \Delta \mathbf{x}$$

- 미분의 Notation을 이렇게 씀.

$$\nabla_{\mathbf{x}} y = \left[\frac{\partial y}{\partial x_1} \quad \cdots \quad \frac{\partial y}{\partial x_D} \right]$$

- 벡터로서, 각 요소가 편미분인 것.

$$\nabla_X f(X) = \left[\frac{\partial f(X)}{\partial x_1} \quad \frac{\partial f(X)}{\partial x_2} \quad \cdots \quad \frac{\partial f(X)}{\partial x_n} \right]$$

- Gradient = $\nabla_x f(x)^T$

Hessian → second derivative of f

$$\nabla^2_x f(x_1,\ldots,x_n) := \begin{bmatrix} \frac{\partial^2 f}{\partial x_1^2} & \frac{\partial^2 f}{\partial x_1 \partial x_2} & \cdots & \frac{\partial^2 f}{\partial x_1 \partial x_n} \\ \frac{\partial^2 f}{\partial x_2 \partial x_1} & \frac{\partial^2 f}{\partial x_2^2} & \cdots & \frac{\partial^2 f}{\partial x_2 \partial x_n} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial^2 f}{\partial x_n \partial x_1} & \frac{\partial^2 f}{\partial x_n \partial x_2} & \cdots & \frac{\partial^2 f}{\partial x_n^2} \end{bmatrix}$$

Minimium of function

> → $f'(x) = 0$인 곳을 찾으면 된다.
> 근데 이게 완벽한 조건은 아님(당연히)

Check the double derivative at x_{soln} : compute

$$f''(x_{soln}) = \frac{df'(x_{soln})}{dx}$$

If $f''(x_{soln})$ is positive x_{soln} is a minimum

- If it is negative, it is a maximum

- 이계도함수를 확인해보면 됨.

Unconstrained Minimization of Function(Multivariate)

- $\nabla_X f(X) = 0$ and
  Hessian Matrix $\nabla_X^2 f(X)$
  - positive → local minima
    negative → local maxima

Approach of Gradient Descent



- derivative의 부호에 따라서
  Step의 방향(-derivative 방향)을 정함 → 최솟값에 가까워지게 됨.

Gradient Descent to train a network

Initialize:

- W^0
- $k = 0$

do

- $W^{k+1} = W^k - \eta^k \nabla Loss(W^k)^T$
- $k = k + 1$

while $\left|Loss\left(W^k\right) - Loss\left(W^{k-1}\right)\right| > \varepsilon$